

Extended Curriculum Vitae

for *Rúben Alexandre Dinis Araújo*

Ph.D. Biomedicine (NOVA Medical School)

MSc. Biomedical Engineering (ISEL)

MSc. Maintenance & Production – Mec. Eng. (ISEL)

BSc. Mechanical Engineering (ISEL)

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Table of Contents

Disclaimer	1
Summary	2
Chapter 1 – Academic Qualifications	3
Chapter 2 – Professional Experience.....	5
Chapter 3 – Technical Skills & Software Proficiency	8
General Computer Software.....	8
Engineering, Data Analysis & Management Software	8
CAD/CAE/CAM.....	8
Calculations.....	8
Statistics	8
Management.....	9
Multivariate Data Analysis / Data Mining	9
Coding.....	9
Spectroscopy	9
Programming Languages.....	9
Chapter 4 – Certifications, Training & Professional Development.....	10
Languages	10
Engineering Design, Simulation & Manufacturing.....	10
Engineering Standards, Quality & Risk Management	11
Programming, Data Analysis & Machine Learning	11
Professional, Compliance, Safety & Corporate Training	12
Additional Certifications & Training	12

Chapter 5 – Funded Projects	13
Chapter 6 – Innovation and Technology Transfer	16
Chapter 7 – Student Supervision, Teaching Activities & Scientific Outreach	17
Student Supervision and Mentoring	17
Teaching Activities.....	17
Scientific Outreach	17
Chapter 8 – Awards & Distinctions.....	19
Chapter 9 – Volunteering & Community Engagement	20

Disclaimer

- 1) Additional evidence, such as academic degrees, diplomas, professional experience, publications, awards, and supporting documentation, is available at: <https://rubenaraujo.org/cv/>.
- 2) While this document is intended to be as complete as possible, it is still not representative of the entire cumulative experience and/or courses taken over the years and, as such, represents a constant work in progress. It is important to note, however, that this particular version of the extended *Curriculum Vitae* is mostly limited to experience in engineering, intellectual property, and any courses that may pertain to these areas. For the complete academic and professional experience, the reader may request, at any time, a version that includes the extensive biomedical research content.

Summary

This version of the *Curriculum Vitae* refers only to professional work developed from 2011 onward, starting from the time the candidate completed his first Master's degree in Mechanical Engineering, in the field of Maintenance and Production, following a Bachelor's degree (2002–2007) in Mechanical Engineering, both from the same technical university (*Instituto Superior de Engenharia de Lisboa* – ISEL). From there onward, experience was accrued at Siemens (*Fábrica do Sabugo*, Sintra, Portugal), and in Norway, both in the oil and gas industry (Aker Solutions, MH Wirth) and in heavy industry and services (Brøvig), before returning to Portugal. There, the candidate successfully completed a second Master's degree (2017–2019), this time in Biomedical Engineering, through a partnership between ISEL and *Escola Superior de Tecnologia da Saúde de Lisboa* (ESTeSL), followed by a short period in the private healthcare industry, before returning to complete his studies with a Doctoral degree at NOVA Medical School (NMS – *Faculdade de Ciências Médicas*, Lisbon, Nov. 2021–Apr. 2026).

Following the **Disclaimer** and this **Summary**, **Chapter 1** details the candidate's complete academic degree information. **Chapter 2** provides a brief summary of all engineering experience from 2011 onward, while **Chapter 3** summarizes the candidate's technical skills and software proficiency in engineering, data analysis, and management, including proficiency levels for each tool. **Chapter 4** lists all certifications obtained by the candidate over the years, both in professional settings and through courses taken at personal expense. While this particular document does not address, as previously mentioned, the research component of the candidate's career, **Chapter 5** still summarizes participation in all funded projects (in table form) within this particular professional setting, which is then complemented in **Chapter 6** with relevant experience in Innovation and Technology Transfer. **Chapter 7** briefly details various subjects, ranging from supervision and mentoring activities to scientific outreach efforts. Chapter 8 presents **Awards & Distinctions**, with the **final chapter** providing examples of some of the volunteering experiences that the candidate participated in over the years.

Keywords: Mechanical Engineering; Maintenance & Production; Industrial Experience; Oil & Gas; Heavy Industry; Biomedical Engineering; Innovation; Technology Transfer; Intellectual Property; Project Management.

Chapter 1 – Academic Qualifications

The following list includes all university-granted academic qualifications, including the respective periods of study and, where applicable, final projects or dissertations. Copies of the corresponding diplomas and certificates are available for personal viewing or record-keeping at: <https://rubenaraujo.org/cv/#education>.

2021 – 2026

NOVA Medical School – <https://www.nms.unl.pt/>
Ph.D. in Biomedicine

Thesis: *Diagnosis and Prognosis Disease Biomarkers on Critically Ill Patients with COVID-19 Towards a Precision Medicine - a Machine Learning Approach.*

Brief description: Research focused on the development of diagnostic and prognostic biomarkers for critically ill COVID-19 patients using Fourier-transform infrared spectroscopy (FTIRS), targeted proteomics, and machine learning. The work demonstrated the potential of biochemical fingerprinting and cytokine profiling for infection detection, infection characterization, and mortality prediction, with models outperforming established ICU severity scores. Additional work addressed delirium biomarker discovery, serum and plasma characterization, and the development of computational tools for biomedical data analysis. Overall, the research emphasized analytical robustness, translational applicability, and the development of accessible data-driven approaches for clinical and research settings.

Identifier: <http://hdl.handle.net/10362/202796>

2017 – 2019

Instituto Superior de Engenharia de Lisboa – <https://www.isel.pt/en/>
MSc. in Biomedical Engineering

Thesis: *Impact of Epigallocatechin-3-gallate (EGCG) on the molecular profile of plasma and serum.*

Brief description: It was investigated the effects of 90-day consumption of epigallocatechin-3-gallate (EGCG), a major green tea catechin, on the molecular profiles of human serum and plasma. Mid-infrared spectroscopy (MIRS) combined with multivariate statistical and machine learning methods was used to identify changes in protein and lipid profiles and potential biomarkers associated with EGCG consumption. The results demonstrated significant molecular profile changes in both serum and plasma, highlighting the potential of MIRS as a rapid, cost-effective, and high-throughput approach for metabolic profiling. The work further supported the application of spectroscopy and automated pattern recognition for large-scale biomedical and epidemiological studies.

Identifier: <http://hdl.handle.net/10400.21/11782>

2007 – 2011

Instituto Superior de Engenharia de Lisboa – <https://www.isel.pt/en/>

MSc. in Mechanical Engineering

Specialization in Maintenance & Production

Thesis: *Vibration Analysis in Reduction Gearboxes.*

Brief description: This Master's research focused on the vibration analysis for condition monitoring and fault diagnosis of industrial gearboxes, comparing a healthy gearbox with one exhibiting gear-pair wear. Vibration signals were analyzed using Fast Fourier Transform (FFT) under varying speed and load conditions. The study demonstrated that load had the strongest influence on vibration amplitudes, particularly at gear-meshing frequencies, harmonics, and sidebands, with more pronounced effects in the worn gearbox. The results reinforced vibration analysis as an effective predictive maintenance tool for reducing downtime and maintenance costs.

Identifier: <http://hdl.handle.net/10400.21/1012>

2002 – 2007

Instituto Superior de Engenharia de Lisboa – <https://www.isel.pt/en/>

BSc. in Mechanical Engineering

Thesis: *Modeling and testing of a two-level automotive elevator.*

Brief description: This Bachelor's thesis focused on the modeling and testing of a two-level automotive lift used in vehicle repair and maintenance facilities. The project involved the complete mechanical (SolidWorks), hydraulic, and control-system modeling of the equipment, comprising more than 1,500 individual components. Structural loads, temperature effects, and other operating conditions were evaluated using COSMOSWorks and MD Solids, while equipment movement was simulated using Visual Nastran.

Identifier: n.a.

Chapter 2 – Professional Experience

The following list summarizes all relevant professional experience, including engineering, biomedical research, and other professional activities. Each entry includes the period of employment, position title, and a brief description of the main responsibilities, projects, or areas of focus.

Nov. 2021 – Oct. 2025 **Fundação para a Ciência e a Tecnologia** – <https://www.fct.pt/>
Lisbon, Portugal
PhD Studentship

Job description: PhD research position focused on biomarker discovery and predictive modeling in critically ill COVID-19 patients. Activities included experimental work, clinical data analysis, machine learning, statistical modeling, scientific writing, development of computational tools supporting biomedical research and public presentations of developed work.

Apr. 2020 – Oct. 2020 **Instituto Superior de Engenharia de Lisboa** – <https://www.isel.pt/>
Lisbon, Portugal
Research Scholarship

Job description: Research position supporting the initial implementation of the PREMO project, a large-scale longitudinal observational study of more than 1,200 ICU patients. Work included development of classification and prediction models using clinical, therapeutic, metabolomic, and proteomic data; serum sample processing and laboratory analysis; data preparation and statistical analysis; operation of laboratory equipment; and training/support of laboratory assistants.

Jan. 2020 – Apr. 2020 **Instituto Português da Face** – <https://www.ipface.pt/>
Lisbon, Portugal
Principal Investigator

Job description: Dedicated to two main projects, BIODISCUS and ARTHUR: 3D Dentofacial Surgery Full Planning, focused on temporomandibular joint tissue regeneration and 3D-based facial surgery planning. Responsibilities included scientific writing, manuscript preparation and submission, peer-review activities, data preprocessing, statistical analysis, and project-related research support.

Oct. 2017 – Apr. 2018 **Memory Particle**
Lisbon, Portugal
CEO / Mechanical Design Lead

Job description: Provided remote engineering support to Brøvig in Norway, including 3D mechanical modeling, preparation and revision of 2D drawings,

engineering database management, equipment updates, development of new production concepts, and technical support to engineers and production teams.

Oct. 2016 – Oct. 2017 **Brøvig** – <https://brovig.com/>
Kristiansand, Vest-Agder, Norway
Lead Mechanical Design

Job description: Responsible for fabrication and assembly drawings for company equipment. Provided shop-floor engineering support, follow-up of equipment through manufacturing and assembly, and mentored colleagues. Activities also included reverse engineering, R&D support for new machinery concepts, management of non-conformities and rework, continuous improvement activities, and 2D/3D plant layout development.

Aug. 2014 – Jun. 2015 **MH Wirth** (now HMW) – <https://hmhw.com/>
Kristiansand, Vest-Agder, Norway
Life Cycle Engineer

Job description: Supported rig managers and service engineers throughout the equipment life cycle. Responsibilities included defining additional work for client projects, performing critical analysis of technical requirements, evaluating maintenance schedules, participating in and leading LCE meetings, monitoring project progress, supporting product owners, coordinating workshop follow-up, and preparing work packages and quotations.

Jun. 2013 – Aug. 2014 **Aker Solutions** – <https://www.akersolutions.com/>
Kristiansand, Vest-Agder, Norway
Project Engineer

Job description: Main designer for the Vertical Pipe Racker (VPR) and associated mechanical, hydraulic, and piping modules. Conducted and supported IAT, FAT, and SAT (Installation, Factory and Site Acceptance Testing, respectively) activities in workshops in Norway and Poland. Applied relevant industry standards, including API, DNV, ABS, ISO, NORSOK, and STATOIL TR (now known as Equinor Technical Requirements). Worked closely with service engineers; mentored and trained junior engineers.

Aug. 2012 – Jun. 2013 **Aker Solutions** – <https://www.akersolutions.com/>
Kristiansand, Vest-Agder, Norway
Product Engineer

Job description: Main designer for Fingerboard systems, with responsibility for Bill of Materials (BOM) definition and technical coordination across internal departments, suppliers, and clients. Worked with structural engineering teams to verify design calculations and interpret hydraulic schematics and Piping and Instrumentation Diagrams (P&IDs) from a mechanical design perspective.

Jun. 2011 – Jun. 2012 **Siemens Energy** – <https://www.siemens-energy.com>
Sabugo, Sintra, Portugal
Mechanical Design Engineer

Job description: Responsible for the mechanical design of Large Distribution Transformers (LDTs) and Distribution Transformer (DT) units, including support for the design of special-purpose ATEX transformers for the petrochemical industry. Responsibilities covered projects from initial concept through Final Inspection and Quality Gates (QGs), including prototype and R&D work for new LDT transformers, finite element analysis (FEA), and thermodynamic studies on novel materials.

Chapter 3 – Technical Skills & Software Proficiency

This chapter summarizes the candidate's technical skills and software proficiency across engineering, data analysis, and management, with proficiency levels provided for each software tool. Programming-language proficiency is also included.

General Computer Software

Microsoft Windows	● ● ● ● ●
Microsoft Word	● ● ● ● ●
Microsoft Excel	● ● ● ● ●
Microsoft Access	● ● ● ○ ○
Microsoft Project	● ● ● ○ ○
GNU/Linux	● ● ● ● ○
macOS	● ● ○ ○ ○

Engineering, Data Analysis & Management Software

CAD/CAE/CAM

SolidWorks	● ● ● ● ●
Creo Elements	● ● ● ● ●
Inventor	● ● ● ● ●
AutoCAD	● ● ● ● ●
Siemens NX	● ● ● ● ○
CATIA	● ● ● ○ ○
Revit	● ● ● ○ ○
Navisworks	● ● ● ○ ○
Plant Design Management System	● ● ● ○ ○

Calculations

ANSYS Workbench	● ● ● ● ○
STAAD.Pro	● ● ● ○ ○
MATLAB	● ● ● ○ ○
Automation Studio	● ● ● ○ ○

Statistics

IBM SPSS Statistics	● ● ● ● ●
GraphPad Prism	● ● ● ● ○
GPower	● ● ● ● ○
Gretl	● ● ● ● ○

Management

SAP	● ● ● ● ●
Teamcentre	● ● ● ● ●
PDMLink	● ● ● ● ○
PTC Windchill	● ● ● ● ○

Multivariate Data Analysis / Data Mining

Anaconda	● ● ● ● ●
Orange 3 data mining	● ● ● ● ●
Power BI	● ● ● ● ○
Tableau	● ● ● ● ○
MS SQL	● ● ● ● ○
PostgreSQL	● ● ● ● ○
DB Browser (SQLite)	● ● ● ○ ○
DB Browser (SQLCipher)	● ● ● ○ ○
SQL Server Management Studio	● ● ● ○ ○
pgAdmin	● ● ● ○ ○

Coding

PyCharm	● ● ● ● ○
Atom	● ● ● ● ○
VS Code	● ● ● ● ●
Thonny	● ● ● ● ●
Git Bash	● ● ● ● ○
GitHub Desktop	● ● ● ● ●
Arduino IDE	● ● ○ ○ ○
RStudio	● ● ● ● ○
Jupyter Lab	● ● ● ● ●
ChatGPT	● ● ● ● ●
Claude	● ● ● ○ ○

Spectroscopy

The Unscrambler X	● ● ● ● ●
OPUS	● ● ● ● ●

Programming Languages

Python	● ● ● ● ○
R	● ● ● ● ○
HTML	● ● ● ○ ○
CSS	● ● ● ○ ○
JavaScript	● ● ● ○ ○
PHP	● ● ● ○ ○

Chapter 4 – Certifications, Training & Professional Development

The following list includes all relevant certifications and courses completed over the years, organized by subject area and covering language proficiency, engineering, programming, technical, and other areas. Each entry includes the course/certification and, where appropriate, a brief description of its content or scope.

Languages

Language	CEFR* Level	Exam	Grade	Certification Body
English	C1	IELTS [°] Academic ^{a)} ESOL ⁺ Level 2 Certificate ^{b)}	8.0/9 B (74/100)	University of Cambridge
Spanish	A2	<i>Certificado de Español</i>	Notable (8/10)	<i>Instituto Cervantes</i>
Norwegian	B1	Certificate of Basic Norwegian (<i>Bokmål</i>)	-	<i>Aftenskolen</i>

* **CEFR**: Common European Framework of Reference for Languages; [°] **IELTS**: International English Language Testing System; ⁺ **ESOL**: English for Speakers of Other Languages; ^{a)} Granted on October 2020; ^{b)} Granted on December 2009

Engineering Design, Simulation & Manufacturing

PTC University – Creo Elements/Parametric 2.0 Training Program

Brief scope: 22-module training program covering parametric and advanced modeling, assemblies, detailing, surfacing, sheet metal, manufacturing, piping, cabling, simulation, mold design, and mechanism design.

Siemens NX 10.0 Essential Training

Siemens NX Unigraphics (UG) – Basic to Advanced

AutoCAD 2023 Essential Training

Autodesk Inventor 2024

SolidWorks 2022

Finite Element Simulations with ANSYS Workbench 2016

Brief scope: Intermediate finite element analysis covering structural optimization, nonlinear materials and simulations, transient structural analysis, and explicit dynamics.

STAAD.Pro V8i Fundamentals

Brief scope: Fundamentals of structural model generation using STAAD.Pro.

Revit MEP 2013

Brief scope: Building Information Modeling (BIM) for mechanical, electrical, and plumbing systems.

Revit Architecture 2015 Essential

Engineering Standards, Quality & Risk Management

HN Metrology Consulting – GPS 100: Geometric Product Specifications

Brief scope: Fundamentals of systematic geometric product specification and tolerancing, including the transition from dimensional to geometrical tolerancing.

ISO 9001:2015 Quality Management System Auditor Course

Brief scope: ISO 9001:2015 requirements, quality management system implementation, internal and external auditing, nonconformities, and corrective actions.

ISO 14971 – Medical Device Risk Management

Brief scope: Medical device risk management, including risk-management processes, Failure Mode and Effects Analysis (FMEA), and Fault Tree Analysis (FTA).

Programming, Data Analysis & Machine Learning

Formabase – Web Creation

Brief scope: Website development and maintenance using HTML, CSS, JavaScript, PHP, and MySQL.

Power BI A–Z: Hands-On Power BI Training for Data Science

Brief scope: Data extraction, transformation, visualization, and reporting using Microsoft Power BI.

SQL & Database Design A–Z: MS SQL Server + PostgreSQL

Brief scope: SQL querying, relational database design, normalization, database management, and OLTP/OLAP (Online Transaction Processing/Online Analytical Processing) concepts using MS SQL Server and PostgreSQL.

Complete Python Bootcamp: From Zero to Hero in Python

Brief scope: Python programming fundamentals, object-oriented programming, modules, exception handling, web scraping, and GUI development.

Data Science A–Z: Real-Life Data Science

Brief scope: Data preparation, visualization, statistical modeling, regression, SQL, ETL (Extract, Transform, Load), data wrangling, and communication of analytical results.

MATLAB Programming for Engineers

Brief scope: MATLAB programming for engineering, mathematics, data analysis, visualization, and numerical problem solving.

Python for Data Science and Machine Learning Bootcamp

Brief scope: Data analysis and machine learning using Python, Pandas, NumPy, SciKit-Learn, regression, decision trees, random forests, support vector machines, neural networks, and Big Data tools.

Complete Guide to TensorFlow for Deep Learning with Python

Brief scope: Neural networks with TensorFlow for classification, regression, time-series analysis, reinforcement learning, convolutional networks, and generative adversarial networks.

Machine Learning A–Z: Hands-On Python & R in Data Science

Brief scope: Machine learning in Python and R, including predictive modeling, dimensionality reduction, reinforcement learning, NLP (Natural Language Processing), and deep learning.

Deep Learning with Python and Keras

Brief scope: Supervised and unsupervised deep learning using convolutional and recurrent neural networks, transfer learning, and GPU/cloud-based model training.

Unsupervised Deep Learning in Python

Brief scope: Unsupervised learning methods including PCA, t-SNE, autoencoders, Restricted Boltzmann Machines, and related neural-network techniques.

Professional, Compliance, Safety & Corporate Training

Siemens – Global Competition

Brief scope: Competition-law principles, restrictive agreements, market dominance, and antitrust compliance.

Siemens – Global Bribery and Corruption Awareness

Brief scope: Identification and prevention of bribery and corruption, applicable compliance principles, and warning signs.

Siemens – Business Conduct Guidelines

Brief scope: Corporate ethics, conflicts of interest, information handling, external-partner relationships, and compliance.

Siemens – Environmental Awareness

Brief scope: Environmental impacts of industrial activities and measures for mitigation and reduction.

Aker Solutions – Health, Safety & Environment (HSE) and Working at Heights

Brief scope: Workplace and test-site safety, hazard identification, nonconformity reporting, risk prevention, and working-at-heights practice.

Additional Certifications & Training

Advanced Open Water Diver

Issued by: PADI 5 Star Training Centre

Emergency First Responder

Issued by: Nautilus-Sub

Chapter 5 – Funded Projects

This chapter summarizes, in table form, all funded projects in which the candidate participated during the Master's in Biomedical Engineering at ISEL, the Ph.D. in Biomedicine at NMS, and as Principal Investigator (PI) at *Instituto Português da Face* (IPF). As this chapter focuses specifically on funded projects, it does not include other projects that either did not require or did not secure funding. The full list and URL links can be found at <https://rubenaraujo.org/cv/#projects>. For improved readability, the table has been moved to the following page, where it is displayed in landscape orientation.

Extended Curriculum Vitae for **Rúben Alexandre Dinis Araújo**

Project Title	Funding Type (value)	Project ID	Brief Description	Role	Institutions Involved *
Arthur: 3D Dentofacial Surgery Full Planning	Grant (479,839€)	POCI-01-0247-FEDER-039690	Development of a solution for aesthetic bone and dental surgery enabling virtual surgical planning and prediction of the impact and final facial outcome of surgical interventions.	Researcher	CODI, IPF, <i>Universidade de Coimbra, Universidade de Aveiro</i>
Predictive Models of COVID-19 Outcomes for Higher Risk Patients Towards a Precision Medicine	Grant (240,000€)	DSAIPA/DS/0117/2020	Development of predictive models for COVID-19 outcomes to support faster and more precise clinical decision-making and reduce severe events, recovery times, mortality, and pressure on specialized healthcare facilities.	Researcher	CHULC, ISEL, IST-ID, CardioID
Biodiscus	Grant (192,730€)	CENTRO-01-0247-FEDER-039969	Design and development of customizable bioactive discs for temporomandibular joint replacement, using 3D printing to replicate the native disc's biological and mechanical properties.	Principal Researcher	IPF, CDRSP
GenTox	Grant (5,000€)	IDI&CA/IPL/2017/GenTox/ESTeSL	Development of a rapid and cost-effective biomonitoring technique to predict exposure to genotoxic agents based on serum molecular profiles.	Researcher	ESTeSL, ISEL
Renal Prognosis	Grant (5,000€)	IDI&CA/IPL/2018/RenalProg/ISEL	Discovery of biomarkers from non-invasive or minimally invasive samples (e.g., serum) to assess renal allograft function and diagnose and predict transplant rejection.	Researcher	ISEL, IPST, CHULC
NephroMD	Grant (5,000€)	IDI&CA/IPL/2020/NephroMD:ISEL	Contribution to the development of a non-invasive wearable sensor for real-time monitoring of the pathophysiological state of patients with Chronic Kidney Disease, particularly during hemodialysis, based on sweat analysis.	Researcher	ISEL, IST-ID

Extended Curriculum Vitae for **Rúben Alexandre Dinis Araújo**

DrugsPlatf	Grant (5,000€)	IDI&CA/IPL/2017/DrugsPlatf/ISEL	Development of a disruptive antibiotic discovery platform based on the rapid identification of mechanisms of action and biological targets.	Researcher	ISEL, ESTeSL
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* **CODI:** Comércio Design Industrial, Lda.; **IPF:** Instituto Português da Face; **CHULC:** Centro Hospitalar Universitário de Lisboa Central; **ISEL:** Instituto Superior de Engenharia de Lisboa; **IST-ID:** Instituto Superior Técnico – Investigação, Desenvolvimento e Inovação em Ciência e Tecnologia; **CDRSP:** Centro para o Desenvolvimento Rápido e Sustentado de Produto; **ESTeSL:** Escola Superior de Tecnologia da Saúde de Lisboa; **IPST:** Instituto Português do Sangue e da Transplantação.

Chapter 6 – Innovation and Technology Transfer

The candidate is a named inventor of the provisional patent code **P2134.1 PP**, with the tentative title of “Computer-Implemented Method for an In Vitro or Ex Vivo Diagnosis or Prognosis of a Sample from a Human Patient and Data Processing Device Thereof”, representative of the work developed on biomarkers in the course of the Doctoral Programme and as the evolution of the work started on the Master’s degree. There was also informal support in the establishment of the international patent **PCT/PT2025/050006** “Allograft Rejection Prognosis”. This is complemented by knowledge of industrial property law and the procedures of relevant regulatory bodies.

Chapter 7 – Student Supervision, Teaching Activities & Scientific Outreach

This chapter briefly summarizes activities involving student supervision, teaching, and scientific outreach, from undergraduate to doctoral level. The complete list is also available at <https://rubenaraujo.org/cv/#research-focused> under “Activities.” These include *pro bono* teaching, mentoring, and the management of IEEE-related initiatives. Other roles, such as class representative and secretary responsible for student bodies across engineering disciplines at ISEL, are not included.

Student Supervision and Mentoring

Over the years, the candidate has provided supervision and support to 9 Master’s theses from ISEL and IST, primarily through the Laboratory of Engineering and Health at ISEL and collaboration protocols between the two institutions. More recently, this supervision extended to 7 Ph.D. students (5 from the Biomedicine Doctoral Programme at NOVA Medical School, 1 from IST, and 1 from NOVA School of Science and Technology). In addition, the candidate has supervised and trained laboratory assistants and undergraduate students in Biomedical and Chemical Engineering, including Erasmus exchange students.

Teaching Activities

Taught laboratory classes at ISEL for Bachelor’s- and Master’s-level students in areas including metabolomics, standard wet-laboratory procedures, sterilization best practices, cleanroom procedures, and cell culture. Other teaching activities included collaborative events with The Lego Group (Ultimate Challenge Week), introducing children to robotics and programming, as well as invited sessions for 12th-grade students at Oeiras International School.

Scientific Outreach

This section includes activities associated with the Institute of Electrical and Electronics Engineers (IEEE), ISEL, and NMS. For an extensive list, including supporting URLs and PDFs, please consult the link provided at the beginning of this chapter. Selected activities include:

- Revival of IEEE student chapter, acting as Chairman (ISEL);
- Creation of EMBS (Engineering, Medicine and Biology Society) chapter in ISEL;
- Welcoming events for Bachelor level classes in Biomedical Engineering;
- Organization of IEEE Leadership camps across the country;
- Participation in the organization body of Biomedical and Health Sciences conferences;
- Regular appearance in interviews on the importance and dissemination of biomedical research;
- Annual conference organization of biomedical engineering students in Portugal;
- Organization of webinars;
- Liaison between student bodies and IEFP/EURES networks;

- Continuous support to Bachelor's and Master's students in preparing CVs and conducting mock interviews;
- Bridge academia and industry to facilitate internships or support for master's theses;
- Session moderator in international conferences;
- Speaker in national and international research conferences;
- Speaker in Big Data and AI in healthcare for international university students.

Chapter 8 – Awards & Distinctions

This chapter presents distinctions and awards obtained during the Master's in Biomedical Engineering and the Doctoral Programme in Biomedicine. This list can be found at <https://rubenaraujo.org/cv/#education> under “Prizes and Awards”, as clickable URLs that lead to downloadable files. These are as follows:

- **First in Class Certificate** (Ph.D. in Biomedicine), with a 19/20 evaluation in the curricular part of the degree;
- **First in Class Diploma** (MSc in Biomedical Eng.), recognizing the highest final course average during the two consecutive years of the degree;
- **Merit Scholarship for Excellence and Academic Achievements**, awarded by *Direção Geral do Ensino Superior* (DGES), for the academic year 2018/2019, for an average of 18.38 out of 20, with the award covering 100% of tuition fees.

Chapter 9 – Volunteering & Community Engagement

This chapter highlights volunteering and community engagement activities, including environmental initiatives, social responsibility projects, *and pro bono* STEM education.

- | | |
|---------------------------------|---|
| Siemens Energy | Participated in reforestation initiatives at multiple sites affected by forest fires in Portugal and in the refurbishment of children's wards at <i>Hospital Santa Maria</i> in Lisbon. |
| Nautilus Sub | Participated in volunteer coastal and underwater clean-up activities in Portugal, including marine debris removal and environmental awareness initiatives involving local communities and diving organizations. |
| The Lego Group / Jump Up | Participated in educational activities introducing children aged 7–15 to robotics and programming through practical exercises, training sessions, and competitions. |